

Pedigree analysis and relatedness inference

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Solutions for exercise set II: Kinship testing

Solution II-1

- a) The marker is an autosomal STR marker. It is autosomal because some of the males are heterozygous, and has too many distinct alleles to be a SNP.
- b) Five alleles (13, 14, 15, 21 and 22) are observed. The allele labels indicate the number of repeats.
- c) Both 4 and 5 have genotype 13/14 (because of their parents).
- d) Individual 3 must have genotype 21/22 (because both 21 and 22 are observed in his children, and neither 21 nor 22 can have come from the mother).
- e) The possible genotypes are 13/13, 13/14, 13/15 and 14/15, each with probability $\frac{1}{4}$.

Solution II-2

- a) Two male individuals are typed, and we are testing paternal half siblings against unrelated. The markers are linked on chromosome 12; their distance is $27.57 - 15.63 = 11.94$ cM.
- b) Without linkage, $LR = 34$. With linkage, $LR = 38.79$, corresponding to a 14% increase.
- c) The *Simple* mutation setting applies an Equal model with mutation rate 0.001. This gives $LR = 40.86$ with linkage, and $LR = 35.84$ if linkage is ignored.

Solution II-3

- a) The case concerns three male individuals, A, B and C. The dispute is whether C is the half brother of A and B (who are undisputed full brothers) or unrelated to them.
- b) There are 48 markers, including 18 linked pairs.
- c) KLINK models linkage between the closest markers on each chromosome.
- d) The total LR is 6.61×10^8 with linkage and 2.25×10^8 without linkage.
- e) Without mutation modelling, the total LR becomes 0. This happens because of the exclusion at marker D2S1360, where the genotypes are incompatible with H1.
- f) (Omitted.)